Internal - I

6) If the transmit power is 1W and carrier freq is 2.4 GHz and the receiver is at a distance of 1 mile (1.6 km) from the transmitter. Assume that the transmitter and receiver antenna gains are 1.6. What is the received power in dB and dBm in the free space of a signal? What is the path loss in dB?

Sln:

Given,

$P_t = 1W$

carrier freq (f) $= 2.4$ GHz

$d = 1.6$ km

$G_t = G_r = 1.6$

To find:

1) P_r in dB and dBm

2) path loss in dB

Received power.

$$P_r = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 d^2}$$

To find λ

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{2.4 \times 10^9} = 0.125m$$

$$= \frac{(1)(1.6)(1.6)(0.125)^2}{16 \times 9.8596 \times (1.6 \times 10^3)^2}$$

$$\boxed{P_r = 9.904 \times 10^{-11} W}$$

P_r in dB

$$(P_r)_{dB} = 10\log(P_r)$$

$$= 10\log(9.904 \times 10^{-11})$$

$$\boxed{(P_r)_{dB} = -100.04 \text{ dB}}$$

dB is represented in 1W

P_r in dBm

Why 30,

$$P_r(dBm) = P_r(dB) + 30$$

1W = 1000 mW

$$= -100.04 + 30$$

$$P(dBm) = 10\log(1000)$$

$$\boxed{P_r(dBm) = -70.04 \text{ dBm}}$$

$$= 10 \times 3$$

$$= 30 \text{ dBm}$$

To convert dB to dBm we add 30

path loss,

$$P_L(dB) = -10\log\left(\frac{G_t G_r \lambda^2}{(4\pi)^2 d^2}\right)$$

$$= -10\log\left(\frac{1.6 \times 1.6 \times (0.125)^2}{16 \times 3.14 \times 3.14 \times (1600)^2}\right)$$

$$= -10\log\left(\frac{0.04}{403849216}\right)$$

$$= -10(-10.0415)$$

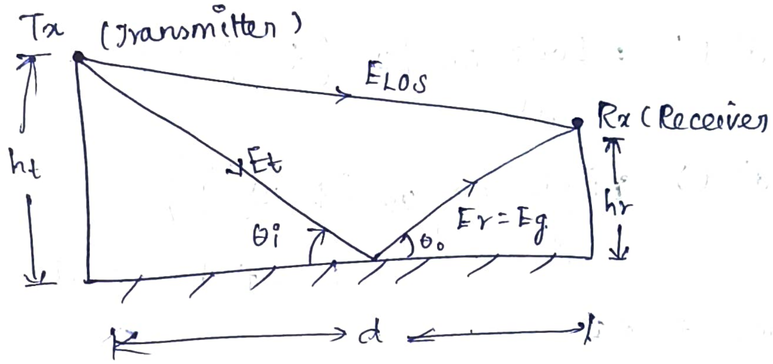
$$\boxed{P_L(dB) = 100.04 \text{ dB}}$$

7) Derive the expression for the received power and path loss in two ray model.

The two-Ray Ground Reflection Model assumes that the received signal consists of two components

1) Direct line of sight (LOS) path

2) Ground reflected path.



$h_t \Rightarrow$ height of the Tx

$h_r \Rightarrow$ " " " Rx

$E_{LOS} \Rightarrow$ E field LOS signal

$E_g \Rightarrow$ E field of Ref signal

$d \rightarrow$ (T-R) separation

Assumptions:

- Flat reflecting ground
- One direct path and one reflected path
- Large distance ($d >> h_t h_r$)
- Reflection coefficient is approximately (-1)

Electric field of LOS signal & reflected signal.

$$E_{LOS}(d',t) = \frac{\cancel{E_0 d_0}}{d'}$$

General exp of E field

$$E(d,t) = \frac{E_0 d_0}{d} \cos\left(\omega_c \left(t - \frac{d}{c}\right)\right)$$

$$E_{LOS}(d',t) = \frac{E_0 d_0}{d'} \cos\left(\omega_c \left(t - \frac{d'}{c}\right)\right)$$

$$E_g(d'',t) = \gamma \frac{E_0 d_0}{d''} \cos\left(\omega_c \left(t - \frac{d''}{c}\right)\right)$$

$\gamma \Rightarrow$ reflection co-eff

The total E field,

$$|E_{tot}| = |E_{LOS}| + |E_g|$$

$$= \frac{E_0 d_0}{d'} \cos\left(\omega_c \left(t - \frac{d'}{c}\right)\right) + \gamma \frac{E_0 d_0}{d''} \cos\left(\omega_c \left(t - \frac{d''}{c}\right)\right)$$

$$|E_{tot}| = \frac{E_0 d_0}{d'} \cos\left(\omega_c \left(t - \frac{d'}{c}\right)\right) - \frac{E_0 d_0}{d''} \cos\left(\omega_c \left(t - \frac{d''}{c}\right)\right)$$

When 'd' becomes large, the difference between d' and d'' is very small,

then,

$$\frac{E_0 d_0}{d} = \frac{E_0 d_0}{d'} = \frac{E_0 d_0}{d''}$$

put $t = \frac{d''}{c}$

$$E_{tot}(d,t) = \frac{E_o d_o}{d}\cos\left(\omega_c\left(\frac{d''}{c} - \frac{d'}{c}\right)\right) - \frac{E_o d_o}{d''}\cos(0)$$

$$= \frac{E_o d_o}{d}\left\{\cos\left(\omega_c\left(\frac{t}{c}\Delta\right)\right) - \cos(0)\right\}$$

By using trignometric identity property,

$$E_{tot} = E_o d_o\left(\frac{4\pi h_t h_r}{\lambda d^2}\right)$$

Receiving power,

$$P_r = E_{tot}^2 \cdot A_e$$

where A_e => Effective Aperture $A_e = \frac{G_r \lambda^2}{4\pi}$

$$P_r = (E_o d_o)^2\left(\frac{4\pi h_t h_r}{\lambda d^2}\right)^2 \cdot \frac{G_r \lambda^2}{4\pi}$$

$$= E_o d_o^2 \frac{(4\pi)^{\cancel{2}}(h_t h_r)^2}{\cancel{\lambda^2} d^4} \frac{G_r \cancel{\lambda^2}}{\cancel{4\pi}}$$

$$= \frac{(P_t G_t)(4\pi)(h_t h_r)^2 G_r}{d^4}$$

$$\boxed{P_r = \frac{P_t G_t G_r (h_t h_r)^2}{d^4}}$$

4π is the constant which is absorbed while deriving the final P_r.

Path loss,

$$P_L = \frac{P_t}{P_r}$$

$$= \frac{\cancel{P_t}}{\dfrac{\cancel{P_t}\, G_t G_r (h_t h_r)^2}{d^4}}$$

$$P_L = \frac{d^4}{G_t G_r (h_t h_r)^2}$$

$$(P_L)_{dB} = 40 \log (d) - \Big\{ 10 \log (G_t) + 10 \log (G_r) + 20 \log (h_t) \cancel{h_r} + 20 \log (h_r) \Big\}$$

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8) ii) Parameters of mobile multiple channels.

9) ii) Parameters of multiple path

Delay spread :

Delay spread is the time difference between the arrival of the first received signal and the last significant multipath signal at the receiver due to multipath propagation.

RMS Delay spread :

RMS (Root Mean Square) Delay spread is the square root of the second central moment of the power delay profile. It measures the time dispersion of multipath signals around the mean excess delay.

Coherance bandwidth :

coherance bandwidth is the range of frequencies over which a wireless ~~com~~ channel has nearly the same response (high correlation).

$$B_c \approx \frac{1}{5\,\tau_{rms}}$$

where,

$B_c \Rightarrow$ coherance Bandwidth

$\tau_{rms} \Rightarrow$ RMS delay spread

Doppler spread :

Doppler spread is the measure of spectral broadening of a signal caused by the relative motion between the transmitter, receiver or surrounding objects.

Coherance time :

Coherance time is the time duration over whi the wireless channel characteristics remain approximately constant.

$$T_c \approx \frac{1}{f_D}$$

where ,

$T_c \Rightarrow$ coherance Time

$f_D \Rightarrow$ Doppler spread

Doppler shift :

Doppler shift is the change in the received signal frequency caused by the relative motion between the transmitter and receiver.

$$f_D = \frac{v}{\lambda} \cos\theta$$

where ,

$f_D \Rightarrow$ Doppler shift

$v \Rightarrow$ Relative velocity

$\lambda \Rightarrow$ wavelength

$\theta \Rightarrow$ Angle between the direction of motion and signal arrival

8) b) Fading due to Multipath time delay spread

Def:

Multipath time delay spread is the time difference between the arrival of the earliest and the latest multipath signals at the receiver.

Causes:

⇒ Reflection from buildings & mountains

⇒ Diffraction around obstacles

⇒ Scattering from trees and vehicles

⇒ Different propagation path lengths

Effects

⇒ Causes Inter symbol Interference (ISI)

⇒ Produces Frequency selective fading

⇒ Distorts the received signal

⇒ Increases the Bit Error Rate (BER)

⇒ Degrades Communication quality

Fading due to Doppler spread

Def:

Doppler spread is the spreading of the received signal frequency caused by the relative motion between transmitter & receiver or surrounding objects

Causes:

⇒ Movement of the mobile user

→ Movement of the Tx or Rx

⇒ " " " surrounding objects (vehicles, people, etc)

Effects:

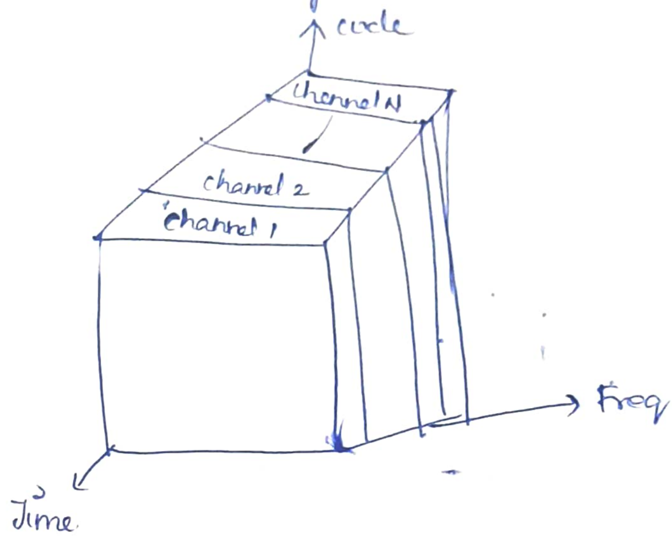
→ Causes Doppler frequency shift

⇒ Produces Fast Fading

⇒ Reduces signal quality

⇒ Increase BER

Q) Explain in detail abt Time Division Multi access technique also give the eq's for capacity & efficiency of TDMA systems.



~~Time~~ TDMA is a digital modulation technique that allows multiple users to share the same freq chanel by dividing the signal to different time slots.

Principle :

⇒ The total available time is divided into small time intervals called time slots.

⇒ A grp of time slots form a TDMA frame

⇒ Each user is allocated one or more time slots in every frame.

⇒ Users transmit only during their assigned slot

⇒ A synchronization system is required to avoid interference b/w users.

eg

⇒ If 4 users share one channel, the frame is divided into 4 time slots

⇒ Users 1 transmits 1 slot, User 2 in slot, and so on.

Operation :

⇒ i/p signals from different users are converted into digital form.

⇒ The signals are stored temporarily in a buffer

⇒ Each user signal is transmitted in its assigned time slot.

⇒ At the Rx side, the cot time slots are seperated & the original signals are recovered.

TDMA frame structure :

1) Guard time ⇒ small time gap between slots to prevent overlapping

2) User data slot ⇒ contains the transmitted info

3) Synchronization bits ⇒ Used to maintain time accuracy.

Capacity eq,

$$C = M \times N$$

$$= \frac{B_{tot} - 2 \cdot B_{guard}}{B_c} \times N$$

$M \Rightarrow$ No. of available radio freq (RF) channel

$N \Rightarrow$ No. of time slots

$B_{tot} \rightarrow$ Total allocated spectrum of the system

$B_c \Rightarrow$ channel bandwidth

Efficiency,

$$\eta = \left(1 - \frac{N b_o + b_g}{B_f}\right) \times 100\,\%$$

$N \Rightarrow$ No. of time slots

$b_o \Rightarrow$ No. of overhead bits per time slot

$b_g \Rightarrow$ No. of equivalent bits in the guard level

$B_f \Rightarrow$ Total no. of bits per frame.

Advantages

⇒ Efficient use of available bandwidth

⇒ No interference b/w users

⇒ provides flexible allocation of time slots

Disadvantages:

• Requires accurate synchronization

⇒ Delay occurs because users must wait for their time slot

⇒ Complex ckt.

Applications:

⇒ Satellite communication

⇒ Wireless " system

⇒ Digital cellular systems

⇒ Digital radio systems

a) a) FDMA, TDMA, CDMA

FDMA	TDMA	CDMA
Users seperated by freq bands	Users are seperated by time slots	Users seperated by unique codes
Available BW is divided into diff freq channels	~~Total ti~~ ~~time slot is~~ Total time is divided into time slots	same freq & time are shared using diff. codes
All users transmit simultaneously but at diff freq's	Users transmit one after another in assigned time slots	All users transmit simultaneously using diff codes
Less efficient	More efficient than FDMA	Highly efficient
No synchronization required	Requires accurate synchronization	Requires code synchronization
Limited no. of users	Higher capacity than FDMA	Higher capacity than three
Simple Imple-mentation	More Complex ckts	High complex ckts
Eg: ~~1G cellular~~ 1G cellular systems	Eg: GSM cellular systems	Eg: 3G Mobile communication system

10)
b)

$B_t = B_{tot} = 12.5\ MHz$

$B_{guard} = 10\ kHz$

$B_c = 30\ kHz$

$$N = \frac{B_t - 2B_{guard}}{B_c}$$

$$= \frac{12.5 \times 10^6 - 2(10^3)}{30 \times 10^3}$$

$$= 416.6$$

$\Rightarrow$ round down becoz 417 channel would exceed the allowed BW limit

$$\boxed{N = 416}$$